

Clinical Guideline for: The management of new onset Atrial Fibrillation (AF) & Atrial Flutter (AFI)

SUMMARY

The purpose of this document is to support safe and effective management of Atrial Fibrillation (AF) and Atrial Flutter (AFI) within the Arrhythmia Care Service for the purpose of rhythm and rate control according to evidence base to include detailed protocols recommended by National Institute for Clinical Excellence (NICE) for patients with AF. As the medical management of Atrial Fibrillation and Atrial Flutter is the same, for the purpose of this document they will both be referred to as AF.

This document covers rate and rhythm control in non-acute settings. Select this link for [rate and rhythm control for people presenting acutely](#) (either new onset or destabilisation of existing atrial fibrillation).

KEY POINTS

- It is not considered exclusive and should be used in conjunction with the British National Formulary and with the Trust's Medicine Policy 2021, and North & East Devon Formulary & Databases.*
- Patients referred to the Arrhythmia service should have a definite diagnosis of atrial fibrillation or atrial flutter confirmed by electrocardiogram.*
- Treatment of new onset AF follows a rate-control or rhythm-control strategy. The aim of treatment is to prevent complications, particularly stroke, and alleviate symptoms. Drug treatments include anticoagulants to reduce the risk of stroke, and an anti-arrhythmic to restore or maintain normal heart rhythm, or to slow the heart rate in those who remain in atrial fibrillation. It also includes the addition of diuretics for patients presenting with symptoms of fluid retention, either due to systolic or diastolic left ventricular dysfunction caused by atrial fibrillation/flutter.*
- All patients referred to the Arrhythmia Service will be assessed with regard to suitability for initiation and optimisation of appropriate beta-blockers and digoxin - in accordance with National Guidance for AF.*
- The Algorithm for the management of persistent atrial fibrillation or atrial flutter, will be utilised to optimise pharmacology for this patient group*
- The Arrhythmia Care Coordinator will obtain the case notes, assess suitability and invite the patient to attend an appointment. During this consultation patients with atrial fibrillation / flutter will be offered a personalised package of care which covers:*
 - stroke awareness and measures to prevent stroke*
 - rate control*

- *assessment of symptoms for rhythm control*
- *who to contact for advice if needed*
- *psychological support if needed*

- *lifestyle advice*
- *up to date and comprehensive education and information on:*
 - *cause, effects and possible complications of atrial fibrillation*
 - *management of rate and rhythm control*
 - *anticoagulation and importance of appropriate control*
 - *support networks*

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1. INTRODUCTION

- 1.1 This clinical guideline is designed to provide advice, guidance and direction to staff whilst leaving room for professional judgement, and adaptation, to fit individual circumstances.**

1.2

Atrial Fibrillation (AF) is the most common heart rhythm disturbance encountered by doctors. NHS 2011 figures suggest AF affects in excess of one million people across the UK, although this is considered to be an under estimation due to delayed detection and diagnosis.

It can affect adults of any age, but it is more common as people get older. In the over 65 year old age group, it affects about 10% of people. Atrial fibrillation is not a life-threatening heart rhythm problem, but it can be troublesome and often requires treatment.

AF occurs when chaotic electrical activity develops in the upper chambers or atria, and completely takes over from the sinus node. As a result, the atria no longer beat in an organised way, and pump less efficiently. The AV node will stop some of these very rapid impulses from travelling to the ventricles, but the ventricles will still beat irregularly and possibly rapidly.

This document is to support safe and effective management of AF within the Arrhythmia Care Service.

2. BACKGROUND

- 2.1** This document was initially designed to support the Arrhythmia Care Coordinators to manage and treat patients with AF. This document replaces the existing guideline which has been updated to include NICE guidelines AF 2021 (replacing NICE guidelines AF 2014) and including the initiation of a diuretic, when indicated.

This document also provides useful guidance to other non-cardiology teams managing AF.

3. DEFINITIONS

- 3.1** Persistent atrial fibrillation/flutter is defined as: “not self-terminating, lasting greater than 7 days or requiring chemical or electrical cardioversion” (Camm et al, 2010).

- 3.2 Resting target heart rate during rate control strategy <110 bpm unless symptomatic, in which case aim for < 80 bpm. In the presence of left ventricular systolic dysfunction, then aim for a resting heart rate of 80 – 90 bpm.

4. MAIN BODY OF GUIDELINE

4.1 Responsibilities

Role of Consultant Cardiologist

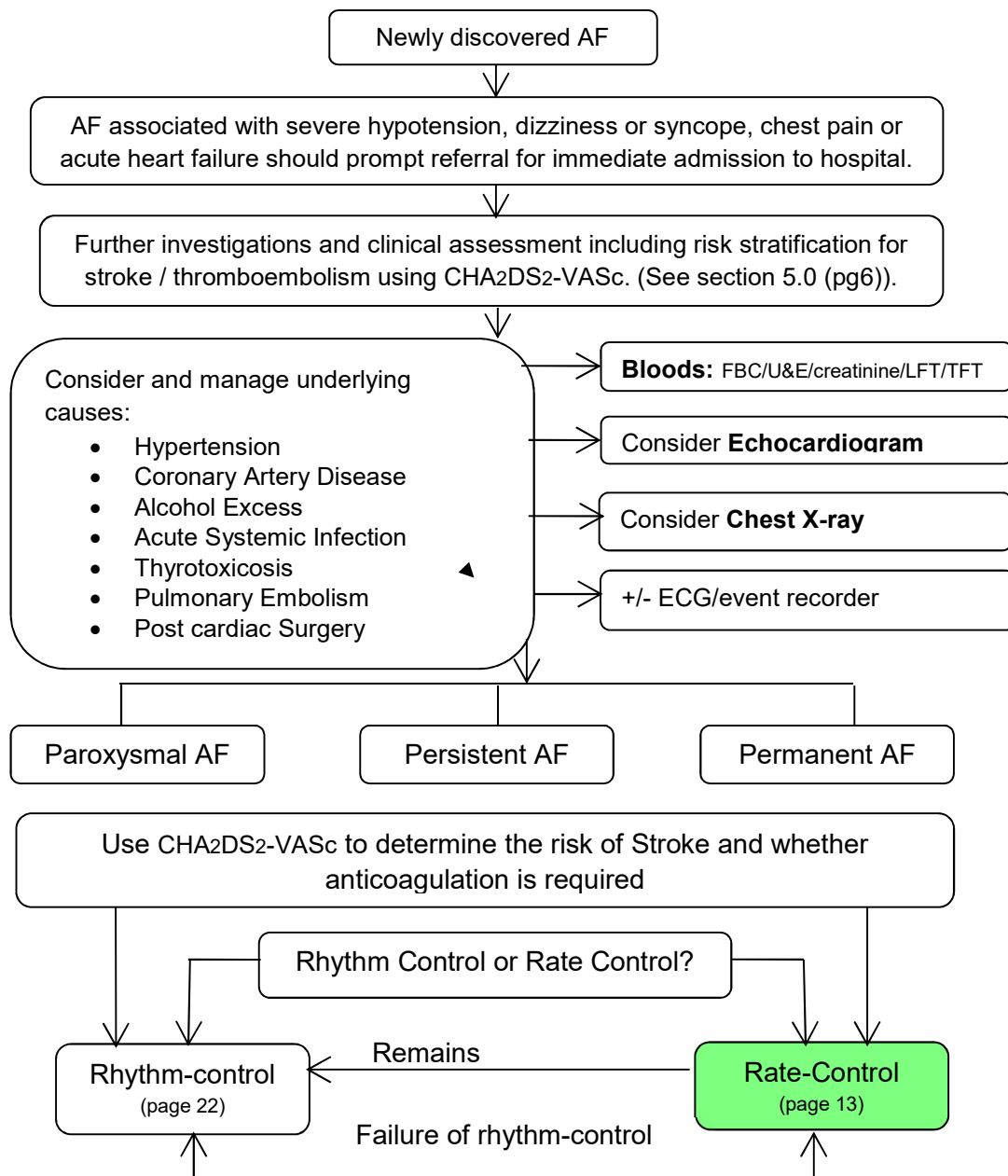
- To retain clinical responsibility for their patient's care while using the arrhythmia service
- To provide clinical support to the arrhythmia care coordinator for the patients being managed within the service

Role of the Arrhythmia Care coordinator

The Arrhythmia Care Coordinator is responsible for:

- Ensuring that the service is delivered adhering to local and national guidelines, ensuring these are regularly reviewed and updated in accordance with the evidence base.
- Ensuring that they work within current legislation and the organisation's Medicine Policy and Formularies.
- Identifying learning needs, meeting their Continuing Professional Development requirements, and ensuring that they are up to date on the conditions which they manage.
- Only practicing within their competence.
- Acting as an expert resource to others (including patients and other professionals).

4.2 Atrial Fibrillation – Treatment Strategy Decision Tree



Offer rate control as the first line treatment strategy for atrial fibrillation except in people:

- whose AF has a reversible cause
- who have heart failure thought to be primarily caused by AF
- with new on-set AF
- with atrial flutter who may be suitable for an ablation strategy
- for whom this strategy is more suitable based on clinical judgement

NICE

NICE AF 2021

4.3 Assessment of Stroke and Bleeding risks

Refer to NICE guidance Atrial Fibrillation: diagnosis and management 2021 NG196

Stroke Risk

Use the CHA₂DS₂-VASc stroke risk score below to assess stroke risk in people with any of the following:

- symptomatic or asymptomatic paroxysmal, persistent or permanent atrial fibrillation
- atrial flutter
- a continuing risk of arrhythmia recurrence after cardioversion back to sinus rhythm.

CHA ₂ DS ₂ -VASc risk factor	Points
Congestive heart failure Signs/symptoms of heart failure or objective evidence of reduced left ventricular ejection fraction	+1
Hypertension Resting blood pressure >140/90 mmHg on at least two occasions or current antihypertensive treatment	+1
Age 75 years or older	+2
Diabetes mellitus Fasting glucose >125 mg/dL (7 mmol/L) or treatment with oral hypoglycaemic agent and/or insulin	+1
Previous stroke, transient ischaemic attack, or thromboembolism	+2
Vascular disease Previous myocardial infarction, peripheral artery disease, or aortic plaque	+1
Age 65–74 years	+1
Sex category (female)	+1

NICE(2014) *Atrial fibrillation: the management of atrial fibrillation - Clinical guideline*
 180

Bleeding risk

Use the [ORBIT score](#) (see below) to assess the risk of bleeding in people who are starting or have started anticoagulation. Advise modification and monitoring of the following risk factors:

- uncontrolled hypertension
- poor control of international normalised ratio (INR) in patients on vitamin K antagonists
- concurrent medication, for example concomitant use of aspirin or a non-steroidal anti-inflammatory drug (NSAID)
- harmful alcohol consumption.
- Reversible causes of anaemia

ORBIT – AF Bleeding Score

O lder Age > 74 years	+1
R educed Haemoglobin/Reduced Haematocrit/Anaemia Hb <130 g/L Male; Hb <120 g/L Females Hct: <40% Males; <30% Females History of anaemia	+2
B leeding risk	+2
I nsufficient Renal Function (eGFR <60 mL/min/1.73m ²)	+1
T reatment with Anti-platelet agents	+1
TOTAL SCORE : 7	

Interpretation

Score	Bleeds per 100 pt years	Bleed Risk Category
0	1.7	Low
1	2.3	
2	2.9	
3	4.7	Medium
4	6.8	High
5	9.0	
6	12.3	
7	14.9	

4.4 Anticoagulation

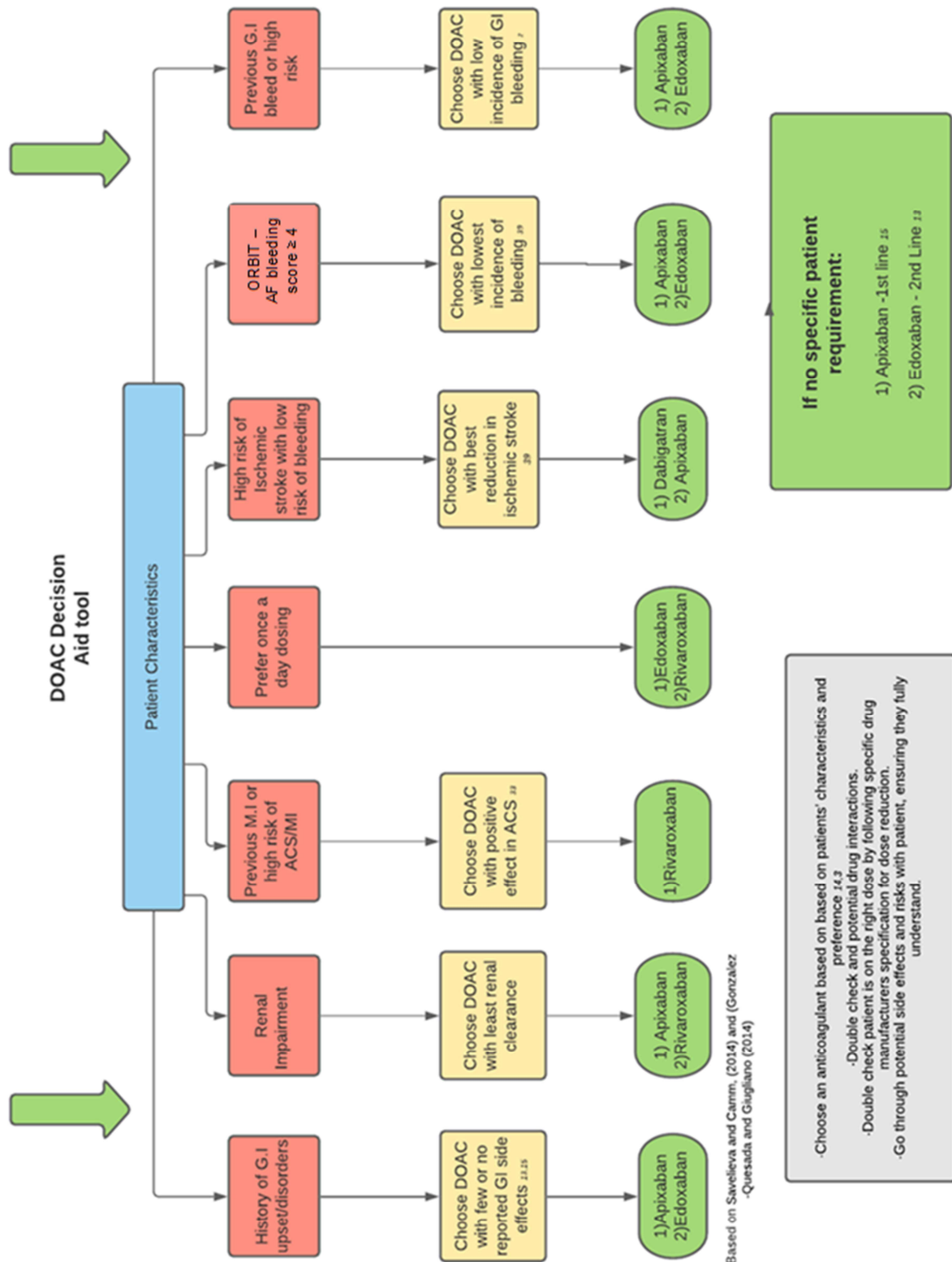
- It is the role of the ACC to consider, or request GPs to consider offering anticoagulation using a direct-acting oral anticoagulant to people with a CHA₂DS₂-VASc score of 1 or above (or 2 for females), taking bleeding risk into account.
- Discuss the risks and benefits of different drugs with the person, taking into account any contraindications for each drug and follow the guidance in the British National Formulary for advice on dosages in people with renal impairment, reversal agents and monitoring.
- When establishing renal function, calculate creatinine clearance using formula with weight and creatinine serum result and NOT eGFR. See <https://www.mdcalc.com/creatinine-clearance-cockcroft-gault-equation>

*Creatinine Clearance = (140 - Age) x wt (kg) x F} / Serum Creatinine (umol).
 (Where F = 1.23 if male and 1.04 if female).*

- Use the CHA₂DS₂-VASc stroke risk score to assess stroke risk in people with a continuing risk of arrhythmia recurrence after cardioversion back to sinus rhythm.
- For adults with atrial fibrillation who are already taking a Vitamin K antagonist and are stable, continue with their current medication and discuss the option of switching treatment, taking into account their time in therapeutic range (TTR).
- Reassess anticoagulation for a person whose anticoagulation is poorly controlled shown by any of the following:
 - 2 INR values higher than 5 or 1 INR value higher than 8 within the past 6 months
 - 2 INR values less than 1.5 within the past 6 months

TTR less than 65%. [2014]
- Do not advise aspirin monotherapy solely for stroke prevention to people with atrial fibrillation.

DOAC Decision Aid Tool



4.5 When to advise rate or rhythm control

This section covers rate and rhythm control in non-acute settings. Select this link for [rate and rhythm control for people presenting acutely](#) (either new onset or destabilisation of existing atrial fibrillation).

Offer rate control as the first-line strategy to people with atrial fibrillation, except in people:

- whose atrial fibrillation has a reversible cause
- who have heart failure thought to be primarily caused by atrial fibrillation
- with new-onset atrial fibrillation
- with atrial flutter whose condition is considered suitable for an ablation strategy to restore sinus rhythm
- for whom a rhythm control strategy would be more suitable based on clinical judgement.

Certain treatments will be discussed with the Consultant Cardiologist, SpR or Staff Grade for Cardiology prior to initiation.

Staff members not holding a non-medical prescriber qualification will request the patients GP to consider initiating / titrating treatment, unless commencing treatment immediately will prevent worsening of the patient's condition which may lead to a hospital admission. In this case, the cardiologist of the week will complete a prescription.

This will be communicated to all relevant parties (GP, hospital personnel involved in continuing care & patient).

Competent Non-medical prescribers will follow these guidelines and communicate prescribing decisions made as above.

4.6 Rate control

In line with North and East Devon Formulary;

- Advise either a standard oral beta-blocker (that is, a beta-blocker other than sotalol) or a rate limiting oral calcium-channel blocker as initial monotherapy to people with atrial fibrillation who need drug treatment as part of a rate control strategy.
- Base the choice of drug on the person's symptoms, heart rate, comorbidities and preferences when considering drug treatment.

- Consider oral digoxin monotherapy for people with non-paroxysmal atrial fibrillation only if they are sedentary (do no or very little physical exercise).
- If monotherapy does not control symptoms, and if continuing symptoms are thought to be due to poor ventricular rate control, as advised by the North & East Devon Formulary & Databases, consider combination therapy with any 2 of the following given orally:
 - beta-blocker OR diltiazem
 - digoxin

or

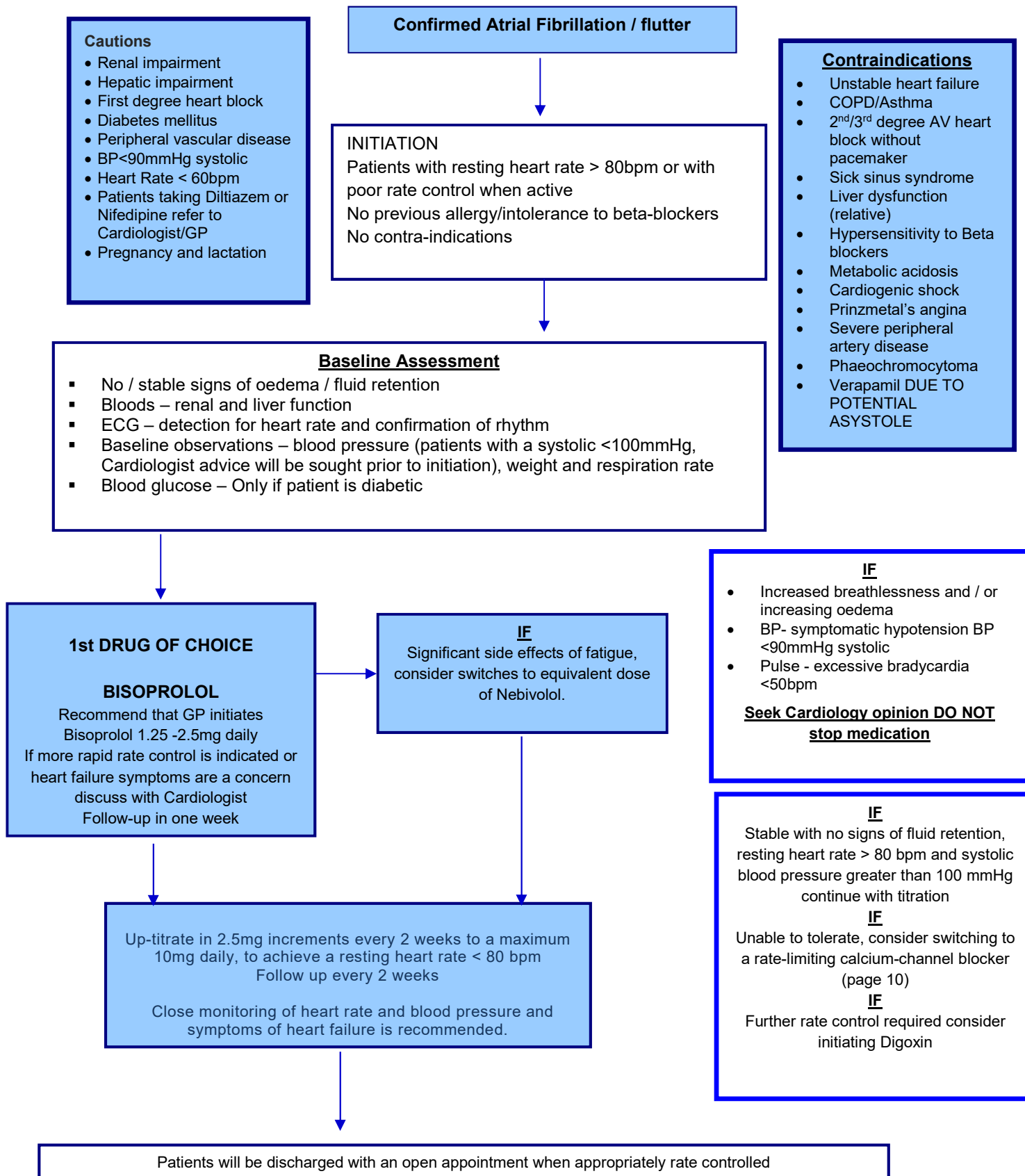
- beta blocker AND diltiazem after consultation with a Cardiologist

Do not offer amiodarone for long-term rate control.

4.7 Beta Blockers

Offer oral Bisoprolol as first line rate control treatment. By inhibiting the cardiac beta receptors, Bisoprolol inhibits the response given to sympathetic activation. This results in a decrease in heart rate and contractility therefore decreasing the oxygen demand of the cardiac muscle.

BETA-BLOCKER Initiation and Titration Algorithm



Advice for patients taking Beta blockers

- Advise that beta-blockers should help to lower heart rate which may improve symptoms.
- Advise to report any other side effects, in particular problems with dizziness / light-headedness / syncope or bronchospasm and fatigue, which could indicate hypotension or bradycardia.
- Advise on other common, potential side-effects i.e. cold peripheries, impotence, sleep disturbances

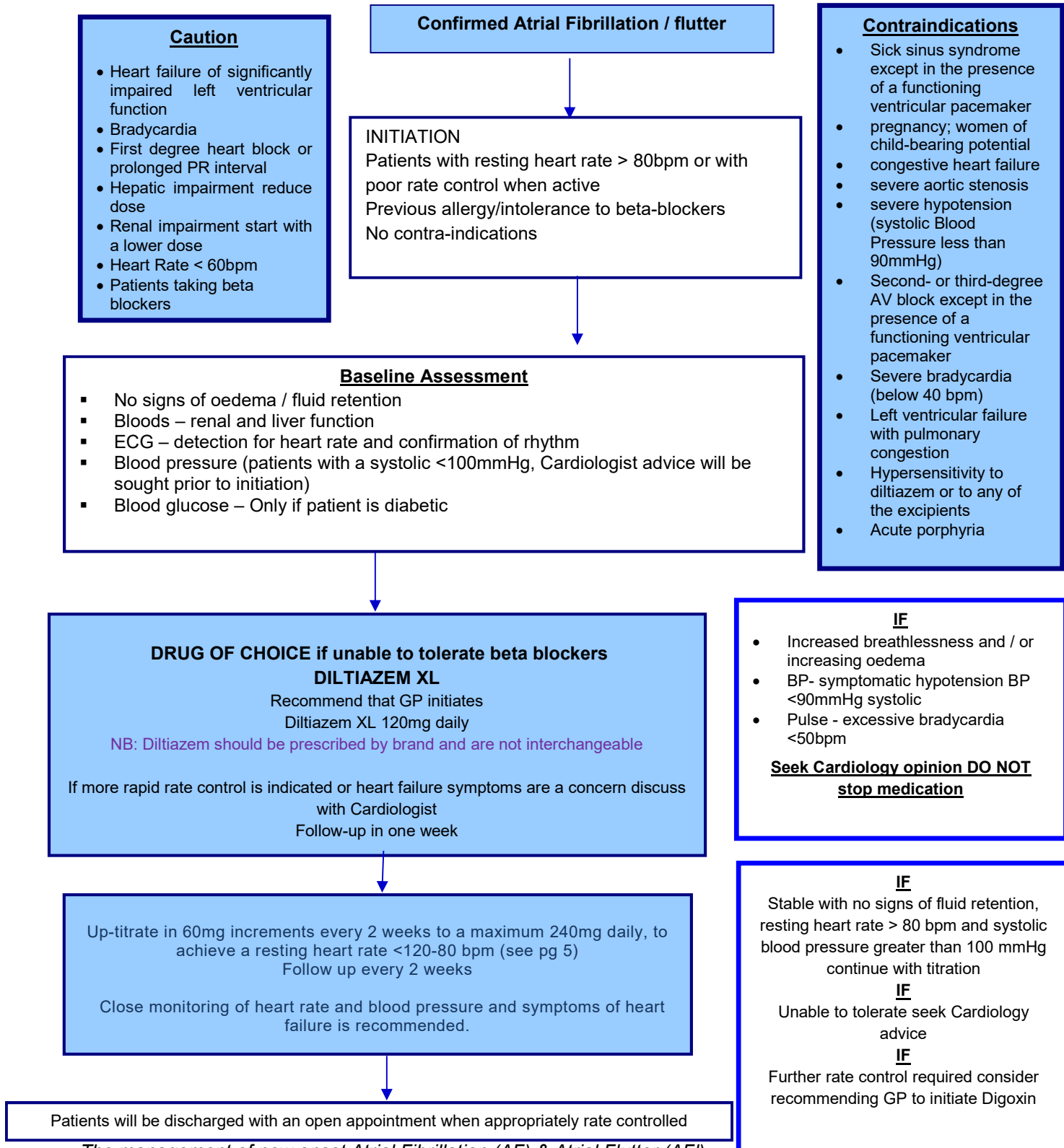
If beta-blocker not tolerated or contraindicated consider a rate-limiting calcium-channel blocker: such as Diltiazem:

Rate Limiting Calcium Channel Blocker

DILTIAZEM

Offer oral Diltiazem hydrochloride if beta blocker is contraindicated or not tolerated. Diltiazem is a calcium antagonist. It selectively reduces calcium entry through voltage-dependent calcium channels into vascular smooth muscle cells and myocardial cells. This lowers the concentration of intracellular calcium which is available to active contractile proteins. In vascular tissue, diltiazem relaxes arterial smooth muscle, reducing systemic peripheral resistance and dilating the coronary arteries. In cardiac muscle diltiazem reduces contractility and slows the heart rate through its negative chronotropic and inotropic actions. Cardiac work and oxygen demand can therefore be reduced and high blood pressure lowered without reflex tachycardia.

Rate-limiting calcium-channel blocker Initiation and Titration Algorithm



- **DIGOXIN**

Offer oral digoxin if further rate control is required when on maximum tolerated dose of either a Beta blocker or rate limiting calcium channel blocker.

Digoxin increases in the intracellular concentration of calcium which, in turn, alters the intracellular disposition of calcium. This results in an increase in the force and velocity of myocardial systolic contraction (i.e. a positive inotropic effect on myocardial cells). Digoxin has a negative chronotropic effect as it slows the overall heart rate. These actions result in a decrease in the rate of contractility, particularly through the atrioventricular node, with subsequent effects on atrial fibrillation. Digoxin also has complex direct effects on conduction velocities and refractory periods of the conducting fibres in cardiac muscle. In addition to its direct effects, digoxin acts indirectly by increasing vagal input through direct vagal stimulation and decreasing sympathetic drive. These effects contribute to the slowing of the ventricular rate by digoxin.

Caution:

Hypokalaemia predisposes to digoxin toxicity; adverse reactions to digoxin may be precipitated if hypokalaemia occurs, for example after prolonged use of diuretics. The effects of digoxin are enhanced by hypokalaemia, hypomagnesaemia, hypercalcaemia, hypoxia, and hypothyroidism and doses may need to be reduced until these conditions are corrected.

Hyperkalaemia occurs in acute digoxin over dosage.

Digoxin doses should generally be reduced and plasma-digoxin concentrations monitored in patients with renal impairment and in the elderly.

(Martindale. 2021. *Medicines complete*. [ONLINE] Available at: <https://www.medicinescomplete.com/> . [Accessed 16 June 2021])

Maintenance dose 125 micrograms daily. Taking into account renal function, age and body weight. Consider:

- 62.5 mcg daily in patients under 50 kg, with stage 3b CKD (eGFR 30-45) or age greater than 70 years

If further ventricular rate control is still required when on maximum tolerated dose of 2 rate controlling agents, discuss with Cardiologist.

Digoxin Initiation and Titration Algorithm

Confirmed Atrial Fibrillation / flutter

CAUTION

- Hypercalcaemia
- Hypokalaemia
- Hypomagnesaemia
- Hypoxia
- Recent myocardial infarction
- Severe respiratory disease
- Sick sinus syndrome
- Thyroid disease
- Constrictive pericarditis
- Hypertrophic cardiomyopathy
- On 2 rate controlling medication

Initiation

Patients with resting heart rate > 110 bpm
 No previous allergy/intolerance to digoxin
 No contra-indications
 Not suitable for initiation or further up-titration of beta-blocker or rate limiting calcium Channel

Contraindications

- Severe renal impairment
- Intermittent complete heart block
- Myocarditis
- Second degree AV block
- Supraventricular arrhythmias associated with accessory conducting pathways e.g. Wolff-Parkinson-White syndrome
- Ventricular tachycardia or fibrillation.

Baseline Assessment

- No signs of oedema / fluid retention
- Bloods – monitor serum electrolytes and renal function
- ECG – detection of heart rate and confirmation of rhythm
- Baseline observations – blood pressure, weight and respiration rate

DRUG OF CHOICE

(if unable to tolerate or on maximum tolerated dose of betablocker or calcium channel blocker)
DIGOXIN

Recommend that GP initiates Digoxin 62.5-125mcg daily.

If more rapid rate control is indicated or heart failure symptoms are a concern discuss with Cardiologist
 Follow-up in two weeks

IF

- Increased breathlessness and / or increasing oedema
- BP- symptomatic hypotension BP <90mmHg systolic
- Pulse - excessive bradycardia <50bpm

Seek Cardiology opinion DO NOT stop medication

IF

Stable with no signs of fluid retention, resting heart rate > 80 110 bpm and systolic blood pressure greater than 100 mmHg continue with titration

Seek Cardiology review

Monitor for signs of toxicity

Note: Digoxin toxicity can occur at any dose of Digoxin, but is more common when the plasma therapeutic concentration is exceeded.

This is usually manifest by anorexia, nausea and vomiting, sometimes xanthopsia (a yellow tint to vision), bradycardia < 60bpm and even ventricular arrhythmias. In elderly patients the symptoms and signs of toxicity may be non-specific, including confusion and falls.

If toxicity is suspected Digoxin should be withheld (at least temporarily) while the patient is investigated urgently. A blood sample should be taken for urgent biochemistry and Digoxin concentration and, if practical, an ECG should be arranged. If toxicity confirmed, the opinion of a cardiologist should sought immediately. Higher serum Digoxin levels increase risk of sudden death.

4.8 Commencement / up-titration of diuretics for patients with fluid congestion due to atrial fibrillation/flutter

Patients with atrial fibrillation/flutter may develop symptoms & signs of fluid congestion, due to either structural or functional cardiac changes resulting in Heart Failure with reduced Ejection Fraction (HFrEF) or Heart Failure with preserved Ejection Fraction (HFpEF), and may require diuretics to relieve breathlessness and oedema.

An echocardiogram and/or NTproBNP should be requested, unless performed in the last 12 months or there are any new changes on a 12-lead ECG, to determine left ventricular systolic and diastolic dysfunction. If systolic dysfunction present, discuss with Cardiologist.

It is the role of the Arrhythmia Care Coordinator to advise the GP of the patient's symptoms and advice if likely to be caused by diastolic dysfunction. The GP will be guided to initiate / up-titrate diuretic therapy as per algorithm, with monitoring of the renal function. It would then be the responsibility of the GP to adjust the dose according to symptoms & signs of congestion, blood pressure and renal function, using the lowest effective dose to maintain euvolemia.

Treatment initiations and titrations will be requested for the GP to consider and initiate, unless commencing treatment immediately will prevent worsening of the patient's condition which may lead to a hospital admission. In which case the cardiologist of the week, or non-medical prescriber, will complete a prescription. This will be communicated to all relevant parties (GP, hospital personnel involved in continuing care & patient).

Advice for patients taking diuretics

- Explain expected benefits: to improve symptoms and exercise capacity.
- Advise patient to take tablets as prescribed, however timing may be altered according to their daily routine, explaining that the diuretic effects will occur within one hour and last for approximately six hours.
- Advise patient to report signs and / or symptoms of fluid congestion i.e. worsening breathlessness, leg swelling, productive cough etc. to GP
- Advise patient to report signs and / or symptoms of dehydration / over-diuresis
- Advise patient to ensure an adequate daily fluid intake; generally 1 ½ - 2 litres (6 – 8 mugs), but to consider an extra 200 - 400 mls (1 – 2 mugs) daily to replace fluid loss during hot weather / fever

Algorithm for the initiation of diuretic therapy for patients with Atrial Fibrillation or Atrial Flutter.

*Alert Levels:

(Before starting treatment)

Urea > 15 mmol/L
 Creatinine > 200µmol/L
 Sodium < 130 mmol/L
 Potassium < 3.5mmol/L
 >5.5mmol/L
 or 30% rise in Creatinine or urea from the baseline

Speak to Cardiologist/GP

Cautions:

- Pregnancy / breast feeding
- Hypotension
- Liver failure
- Prostatic enlargement
- Porphyria
- Already on other

Baseline U&E

Assessment of patient for signs of sodium and water retention:

- Pitting oedema e.g. ankle, leg and sacral area etc
- Pulmonary oedema e.g. chest x-ray or Crackles
- Increasing shortness of breath
- Raised JVP (jugular venous pressure)
- Sudden weight gain (more than 1-2 kg in less than 3 days)

Are there any of the above?

YES

1st Choice - Furosemide
 Consider modest dose 20 - 40mg

Is the patient already on a loop-diuretic?

NO

GP advised to start patient on Furosemide

YES

GP to increase dose of loop diuretic

GP to adjust dose according to symptoms, signs of congestion, blood pressure and renal function.

ADVERSE EFFECTS / SIGNS OF DEHYDRATION:

- Dizziness
- Constipation
- Weight loss >1kg/day
- Reduced skin turgor
- Disproportionate rise in urea
- Fatigue
- Thirst
- Oliguria

Additional clinical signs and symptoms of electrolyte imbalance / dehydration:

- Weakness, lethargy, drowsiness
- Collapse, restlessness
- Muscle pains / cramps, muscle fatigue
- Hypotension
- Tachycardia
- Gastrointestinal disturbances such as nausea, vomiting
- Incontinence
- Gout can also be considered a sign

4.9 Rhythm control

- Consider pharmacological and/or electrical rhythm control for people with atrial fibrillation whose symptoms continue after heart rate has been controlled or for whom a rate-control strategy has not been successful.

Refer to the DC cardioversion policy which can be found on the Trusts intranet site.

- Assess the need for drug treatment for long-term rhythm control, taking into account the person's preferences, associated comorbidities, risk of treatment and likelihood of recurrence of atrial fibrillation.
- If drug treatment for long-term rhythm control is needed, consider a standard beta-blocker (that is, a beta-blocker other than sotalol) as first-line treatment unless contraindications.

For dose initiation, titration and contraindication see rate control.

If beta-blockers are contraindicated or unsuccessful, discuss with Cardiologist.

- Consider amiodarone for people with left ventricular impairment or heart failure, discuss with Cardiologist.
- In people with paroxysmal atrial fibrillation (not flutter), a 'pill in the pocket' strategy for the use of Flecainide should be considered for those who:
 - have no history of left ventricular dysfunction, valvular or ischaemia heart disease and
 - have a history of infrequent symptomatic episodes of paroxysmal atrial fibrillation and
 - have a systolic blood pressure greater than 100 mmHg and a resting heart rate >70 bpm and
 - are able to understand how to, and when to, take the medication

Discuss with Cardiologist

4.10 Left Atrial Ablation and Pace and Ablate Strategy

Left atrial ablation – discuss with Cardiologist

If drug treatment has failed to control symptoms of atrial fibrillation or is unsuitable:

- offer left atrial catheter ablation to people with paroxysmal atrial fibrillation
- consider left atrial catheter or surgical ablation for people with persistent atrial fibrillation
- discuss the risks and benefits and take into account the patients preferences. Explain the procedure is not always effective and that the resolution of symptoms may not be long-lasting

Consider left atrial surgical ablation at the same time as other cardiothoracic surgery for people with symptomatic atrial fibrillation

Pace and ablate strategy – discuss with Cardiologist

Consider pacing and atrioventricular node ablation for people with permanent atrial fibrillation with symptoms or left ventricular dysfunction thought to be caused by high ventricular rates.

When considering pacing and atrioventricular node ablation, reassess symptoms and the consequent need for ablation after pacing has been carried out and drug treatment further optimised.

Consider left atrial catheter ablation before pacing and atrioventricular node ablation for people with paroxysmal atrial fibrillation or heart failure caused by non-permanent (paroxysmal or persistent) atrial fibrillation.

4.11 References

European Society of Cardiology (ESC) Guidelines for the diagnosis and management of atrial fibrillation -2020

NICE clinical guideline 196 - Atrial Fibrillation: diagnosis and management – April 2021

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5. MONITORING COMPLIANCE WITH THIS GUIDELINE

5.1 Standards/ Key Performance Indicators

Key performance indicators comprise:

- *Patients with AF are managed according to this policy*
- *Patients are assessed for stroke risk and offered appropriate tailored treatment*

5.2 Process for Implementation and Monitoring Compliance and Effectiveness

The policy will be implemented by the Arrhythmia Care Coordinator (ACC), following approval from the Cardiology Operational and Governance Group, and Medicines Group.

Monitoring of this policy will be the responsibility of the ACC ensuring:

- *stroke prevention and crisis aversions are reported daily*
- *medication and alterations are collated daily*
- *compliance is documented in the ACC activity record which is reported monthly to the lead nurse of the Cardiac Support Services*
- *when non-compliance is identified, support and advice will be provided to improve practice.*

6. ASSOCIATED CLINICAL GUIDELINES OR POLICIES/PROCEDURES

7. PUBLICATION DETAILS

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